Central Venous Catheter-Related Deep Vein Thrombosis

Objective:

To provide guidance on the diagnosis, treatment, and prevention of central venous catheter-related deep vein thrombosis (DVT).

Background:

Central venous catheters are widely used for resuscitation; administration of chemotherapy, antibiotics and other medications; transfusion therapy; apheresis procedures; dialysis; parenteral nutrition; blood sample acquisition; and supportive care, especially in patients with malignancy. The incidence rates for central venous catheter-related DVT vary widely among studies (ranging from approximately 2 to 70%), due in part to variable criteria for investigation (eg, symptomatic patients versus routine screening of all patients with central venous catheters), as well as differences in patient populations (eg, cancer, surgical, pediatrics) and duration of follow-up. Risk factors for catheter-related DVT include use of larger diameter catheters with multiple lumens, catheter tip malposition, left-sided placement, concomitant infection, history of DVT, and, in patients with cancer, the type and stage of cancer, and administration of chemotherapy. Peripherally inserted central catheters (PICCs) are associated with a higher risk of DVT than other centrally inserted catheters, including implanted ports. Heritable thrombophilias likely increase the risk, however the magnitude of risk increase is unclear and screening for thrombophilia is not currently indicated when considering central catheter placement.

Diagnosis of Catheter-Related DVT:

Patients with central venous catheter-related DVT may develop unilateral hand or arm swelling, pain or swelling in the neck or shoulder, visible collateral veins on the chest, or symptoms of superior vena cava obstruction, such as facial swelling, dyspnea, and headache. Patients with central venous catheter-related DVT may also be asymptomatic. Symptoms of embolization (eg, pulmonary embolism, paradoxical stroke) should also prompt an evaluation for DVT in patients with central venous catheters. While a combination of clinical decision rules with D-dimer can be effective ruling out DVT in certain populations, its efficiency is limited in those with central venous catheters and is not recommended. The best initial test for diagnosing catheter-related DVT is duplex ultrasonography. Isolated subclavian vein or more central venous thrombosis may not be well seen with ultrasonography due to anatomical hindrance by the clavicle and chest wall. In difficult cases, magnetic resonance imaging (MRI) or computed tomography (CT) may be required. Direct contrast venography is an invasive procedure and may be difficult to obtain.

Treatment of Catheter-Related DVT:

The goals of treatment for catheter-associated DVT are to improve acute symptoms, decrease long-term morbidity, prolong patency and survival of the catheter, and prevent embolization and recurrent thrombosis. Acute treatment recommendations are based primarily on trials in patients with lower extremity DVT. Removal of the central venous catheter is not required if it is still needed, functioning properly, and not associated with infection; however, if symptoms persist or worsen despite anticoagulation, the central venous catheter may need to be removed.

A systematic review and meta-analysis of the literature assessing the management of catheter
extremity DVT has shown that low molecular weight heparin (LMWH) and direct oral anticoagu

effective and safe. Longer-term anticoagulation may involve continuation of LMWH alone, LMV
warfarin, or DOACs. If LMWH is transitioned to warfarin, there should be an overlap for a mini
until the international normalized ratio (INR) is therapeutic. Once treatment is initiated, the 3-m
venous thromboembolism (VTE) and major bleeding complication was reported at 0.6% and 2.3%, respectively.

Although treatment duration for catheter-related DVT is controversial, it is reasonable to treat patients with a DVT in the axillary or a more proximal upper extremity deep vein for a minimum of 3 months or longer if the catheter remains in place. In those with a DVT involving only the brachial vein or thrombosis confined to the superficial veins, such as the cephalic or basilic vein, treatment with anticoagulation has not been studied. In this situation, anticoagulation with either full dose or lower dose of anticoagulation (eg, LMWH or DOAC) to prevent progression of the thrombus while the catheter remains in place may be reasonable on a case-by-case basis. There is limited data to guide the use of catheter-directed thrombolysis in patients with extensive central catheter-related DVT and so the use of this intervention remains a case-by-case decision, best guided by consultation with a thrombosis expert and/or interventional radiologist.

Prevention of Catheter-Related DVT:

The most important aspects of prevention of catheter-related DVT include use of central venous catheters only when necessary, insertion of the smallest catheters that satisfy their purpose, and prompt removal when no longer needed. The process of PICC exchange has recently been identified as a risk factor for developing DVT, so should be done only when absolutely necessary.

Routine use of thromboprophylaxis in patients with central venous catheters with anticoagulation is not recommended by most guidelines. A systematic review and meta-analysis evaluating anticoagulation thromboprophylaxis (DOACs, LMWH, warfarin) in patients with cancer and central venous catheters found that thromboprophylaxis compared to no prophylaxis probably decreases the risk of symptomatic catheter-related VTE by approximately 40-50%, with no difference in major bleeding or mortality. Low dose apixaban and rivaroxaban have been shown to be effective as thromboprophylaxis in ambulatory patients with cancer who are at high risk for VTE (Khorana score ≥2), but these studies did not specifically assess those with central venous catheters. A pilot study has been completed with a full-scale study (TRIM-Line trial) underway to compare low dose rivaroxaban to placebo as thromboprophylaxis in patients with cancer and central venous catheters. Apixaban was assessed in children with acute lymphoblastic leukemia or lymphoma and a central venous catheter. No significant benefits were identified in patients receiving apixaban and bleeding complications were infrequent but higher in those receiving apixaban. Hence, thromboprophylaxis should only be use in pediatric patients in whom the potential benefits are thought to outweigh the risk of bleeding.

Thromboprophylaxis may be considered in patients with cancer when the perceived risk of thrombosis outweighs the risk of bleeding and the burden of anticoagulation (e.g., in those with prior venous thrombosis, high Khorana score, etc.).

Pediatrics:

Central venous catheters are a frequent necessity in children who require supportive care to manage their illness (eg, chemotherapy, parenteral nutrition, antibiotics, transfusions). The incidence of thrombosis related to catheter use in children is estimated at 3% to 34% of children with central venous catheters, varying between patient populations and with different diagnostic modalities. Without a previous history of thrombosis, thromboprophylaxis in children with central venous catheters is not routinely recommended but could be considered in children with long-term VTE risk including those on home total parenteral nutrition or undergoing hemodialysis. In children with cancer without previous VTE, central venous catheters alone are not an indication for thrombo

other VTE risk factors are present (including asparaginase chemotherapy, obesity, hormonal c
adolescence, or hospitalization for surgery), thromboprophylaxis with LMWH could be conside

LMWH and warfarin remain the mainstay of the treatment for catheter-related DVT in pediatrics. Both dabigatran and rivaroxaban have been compared to standard of care as treatment of VTE in randomized controlled trials in the pediatric population, which included catheter-related DVT, demonstrating similar efficacy and bleeding risk.

Pediatricians with expertise in thromboembolism should be involved in decisions about thrombosis prophylaxis and management of thromboembolism in pediatric patients where possible. When this is not possible, a combination of a neonatologist/pediatrician and an adult hematologist, supported by consultation with an experienced pediatric hematologist, is recommended.

Other Relevant

- Cancer and Thrombosis
- Deep Vein Thrombosis: Treatment

s clinical guides remain current and reliable. Your support helps protect the quality and continuity of these resources for clinicians nationwide.

References:

Apixaban for the acute treatment of venous thromboembolism in children. Available from: https://clinicaltrials.gov/ct2/show/NCT02464969.

Brandt W, et al. Efficacy and safety of apixaban for primary prevention of thromboembolism in patients with cancer and a central venous catheter: A subgroup analysis of the AVERT Trial. Thromb Res 2022;216:8-0.

Carrier M, et al. Apixaban to prevent venous thromboembolism in patients with cancer. N Engl J Med 2019;380:711-719.

Chopra V, et al. Risk of venous thromboembolism following peripherally inserted central catheter exchange: an analysis of 23,000 hospitalized patients. Am J Med 2018;131(6):651-660.

Davies GA, et al. A prospective study of rivaroxaban for central venous catheter associated upper extremity deep vein thrombosis in cancer patients (Catheter 2). Thromb Res 2018;162:88-92.

Debourdeau P, et al. International clinical practice guidelines for the treatment and prophylaxis of thrombosis associated with central venous catheters in patients with cancer. J Thromb Haemost 2013;11(1):71-80.

Giordano P et al. Recommendations for the use of long-term central venous catheter (CVC) in children with hemato-oncological disorders: management of CVC-related occlusion and CVC-related thrombosis. On behalf of the coagulation defects working group and the supportive therapy working group of the Italian
of Pediatric Hematology and Oncology (AIEOP). Ann Hematol 2015;94(11):1765-1776.

Halton J, et al. Dabigatran etexilate for the treatment of acute venous thromboembolism in chil
randomised, controlled, open-label, phase 2b/3, non-inferiority trial. Lancet Haematol 2021;8(1

Hokusai study in pediatric patients with confirmed venous thromboembolism (VTE). Available from: https://clinicaltrials.gov/ct2/show/NCT02798471.

Ikesaka R, et al. Thromboprophylaxis with rivaroxaban in patients with malignancy and central venous lines (TRIM-Line) a two-center open-label pilot randomized controlled trial. Res Pract Thromb Haemost 2021;5(4):e12517.

Jaffray J, Goldenberg N. Current approaches in the treatment of catheter-related deep venous thrombosis in children. Expert Rev Hematol 2020;13(6):607-617.

Kahale LA, et al. Anticoagulation for people with cancer and central venous catheters. Cochrane Database Syst Rev 2018;6:pCD006468.

Khorana AA, et al. Rivaroxaban for thromboprophylaxis in high-risk ambulatory patients with cancer. N Engl J Med 2019;380(8):720-728.

King HL, et al. Unveiling the complexities of catheter-related thrombosis: risk factors, preventive strategies, and management. J Thromb Thrombolysis. 2025 Mar 5. doi: 10.1007/s11239-025-03073-7.

Kovacs MJ. Central venous catheter (CVC) associated deep vein thrombosis (DVT) in cancer patients (NCT03100071). June 2, 2021]; Available from: https://clinicaltrials.gov/ct2/show/NCT03100071.

Kovacs MJ, et al. A pilot study of central venous catheter survival in cancer patients using low-molecular-weight heparin (dalteparin) and warfarin without catheter removal for the treatment of upper extremity deep vein thrombosis (The Catheter Study). J Thromb Haemost 2007;5(8):1650-1653.

Li A, et al. Efficacy and safety of primary thromboprophylaxis for the prevention of venous thromboembolism in patients with cancer and a central venous catheter: A systematic review and meta-analysis. Thromb Res 2021;208:58-65.

Male C, et al. Rivaroxaban compared with standard anticoagulants for the treatment of acute venous thromboembolism in children: a randomised, controlled, phase 3 trial. Lancet Haematol 2020;7(1):e18-e27.

Monagle P, et al. Antithrombotic therapy in neonates and children: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141(2 Suppl):e737S-801S.

O'Brien SH, et al. Apixaban versus no anticoagulation for the prevention of venous thromboembolism in children with newly diagnosed acute lymphoblastic leukaemia or lymphoma (PREVAPIX-ALL): a phase 3, open-label, randomised, controlled trial. Lancet Haematol 2024;11(1):e27-e37.

Paglialonga F, et al. Vitamin K antagonists in children with central venous catheter on chronic haemodialysis: a pilot study. Pediatr Nephrol 2016;31(5):827-32.

Ruud E, et al. Low-dose warfarin for the prevention of central line-associated thromboses in children with malignancies--a randomized, controlled study. Acta Paediatr 2006;95(9):1053-1509.

Tullius BP, et al. The identification of at-risk patients and prevention of venous thromboembolism in pediatric cancer: guidance from the SSC of the ISTH. J Thromb Haemost 2018;16(1):175-180.

van Es N, et al. A clinical decision rule and D-dimer testing to rule out upper extremity deep vein thrombosis in high-risk patients. Thromb Res 2016;148:59-62.

Wang TF, et al. Management of catheter-related upper extremity deep vein thrombosis in patients with cancer: a systematic review and meta-analysis. J Thromb Haemos 2024;22(3):749-764.

Zwicker JI, et al. Catheter-associated deep vein thrombosis of the upper extremity in cancer pa
from the SSC of the ISTH. J Thromb Haemost 2014;12(5):796-800.

Date of version: 10April2025

Please note that the information contained herein is not to be interpreted as an alternative to medical advice from your doctor or other professional healthcare provider. If you have any specific questions about any medical matter, you should consult your doctor or other professional healthcare providers, and as such you should never delay seeking medical advice, disregard medical advice or discontinue medical treatment because of the information contained herein.